



SYDNEY BOYS HIGH
MOORE PARK, SURRY HILLS

December 2009
ASSESSMENT # 1
YEAR 11

Mathematics Extension

General Instructions:

- Reading time—5 minutes.
- Working time—90 minutes.
- Write using black or blue pen.
- Board approved calculators may be used.
- All necessary working should be shown in every question if full marks are to be awarded.
- Marks may NOT be awarded for messy or badly arranged work.
- Answer in simplest exact form unless otherwise stated.

Total marks—80 Marks

- Attempt ALL questions.
- The mark-value of each question is boxed in the right margin.
- Start each NEW question in a separate answer booklet.
- Hand in your answer booklets in 4 sections: Q1, Q2, Q3, and Q4.

Examiner: Mr D. Hespe

This is an assessment task only and does not necessarily reflect the content or format of the Higher School Certificate.

Question 1 (20 marks)

- (a) Express $\frac{2}{3x+6} + \frac{3x+2}{2x^2+x-6}$ as a simplified fraction. 2
- (b) Solve the following for x :
- (i) $7 - 2x > 12$, 1
- (ii) $|3x - 2| = 4$, 2
- (iii) $x^{5/3} = 32$, 1
- (iv) $\frac{1}{x} \geq \frac{1}{6}$. 2
- (c) If x degrees is equal to X radians, find the value of $\frac{x}{X}$ to two decimal places. 1
- (d) If $\tan \frac{\theta}{2} = t$, express $\cot \theta$ in terms of t . 1
- (e) The normals to the parabola $x^2 = 4ay$ at the points $P(2at, at^2)$ and $Q(2as, as^2)$ meet at R .
- (i) Find the coördinates of R in terms of s and t . 4
- (ii) If $st = -2$, find the cartesian equation of the locus of R . 3
- (f) A curve is given parametrically by the equations $x = t^2$, $y = t^3$, where t is a parameter. 3
- Show that the tangent at the point with parameter t has the equation $2y - 3tx + t^3 = 0$.

Question 2 (20 marks)

(Use a separate writing booklet.)

- (a) P is the point $(\cos \alpha, \sin \alpha)$ and Q is the point $(\cos \beta, \sin \beta)$ on the circumference of the circle, centre the origin, radius 1. By writing down two different expressions for PQ^2 , deduce the result that

3

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta.$$

- (b) Factorise the expression $x^2 - 6x + 9$, and hence find any real solutions for $x^4 - x^2 + 6x - 9 = 0$.

3

- (c) The equation $\frac{5}{y^2} - \frac{1}{2y} = 1$ is true when $y = 2$.

2

Use this fact to find a solution of $\frac{5}{(2y+5)^2} - \frac{1}{2(2y+5)} = 1$.

- (d) Find the acute angle between the curves $y = x^2$ and $y = 3 - 2x$ at the point where they meet in the first quadrant.
(Give your answer to the nearest degree.)

3

- (e) If α , β and γ are the roots of the equation $x^3 + 2x^2 + 3x + 4 = 0$, find the value of

(i) $(\alpha - 1)(\beta - 1)(\gamma - 1)$,

3

(ii) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$,

1

(iii) $\alpha^2 + \beta^2 + \gamma^2$.

1

- (f) (i) Find the value of n for which the division of $3x^n - 7x^2 + 15$ by $x - 2$ leaves a remainder of 11.

2

- (ii) Given that $16x^4 - 4x^3 - 4b^2x^2 + 7bx + 18$ is divisible by $2x + b$, show that $b^3 - 7b^2 + 36 = 0$.

2

Question 3 (20 marks)

(Use a separate writing booklet.)

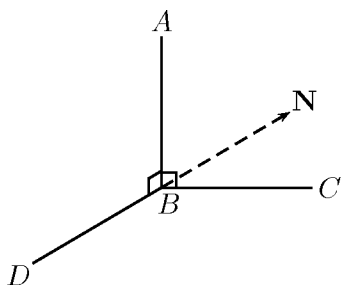
- (a) The points $P(-5, 1)$, $A(a, -3)$ and $B(-2, b)$ are such that P divides the interval AB externally in the ratio $4 : 1$. Find the values of a and b .

3

- (b) Solve $(x + \frac{1}{x})^2 + 6 = 5(x + \frac{1}{x})$ for x .

3

(c)



B , C and D are three points on a horizontal plane.

C is due east of B and D is due south of B .

C and D are 400 metres apart.

A is the top of a vertical tower AB .

From C the angle of elevation of A is 35° and from

D the angle of elevation of A is 50° .

The height of the tower is h metres.

- (i) Copy the diagram and mark on it the given distances and angles.

1

- (ii) Show that $\frac{h^2}{\tan^2 50^\circ} + \frac{h^2}{\tan^2 35^\circ} = 400^2$.

2

- (iii) Find h , correct to one decimal place.

1

- (iv) Find the bearing of D from C to the nearest degree.

2

- (d) (i) Find the set of values of x for which $\frac{x^2 - 12}{x} > 1$.

3

- (ii) Deduce the set of values of x for which $\frac{x^2 - 12}{|x|} > 1$.

1

- (e) (i) For what values of x between 0 and 360 is $\cos x^\circ - \sqrt{3} \sin x^\circ$ positive?

2

- (ii) Find all values of ψ for which $\tan^2 \psi = \tan \psi$.

2

Question 4 (20 marks)

(Use a separate writing booklet.)

- (a) Consider the curve expressed parametrically as $x = t + 1$, $y = 2t^2 - 2$.
- (i) Re-express this relationship in Cartesian form, *i.e.*, as $y = f(x)$. 1
- (ii) Find its focus and directrix. 2
- (iii) Derive the equation of the tangent at the point $A(x_1, y_1)$ on the curve. 2
- (iv) Hence or otherwise, show that the chord of contact of the tangents from an external point $T(x_0, y_0)$ is $y + y_0 = 4(xx_0 - x - x_0)$. 4
- (b) Find the general solution to $\tan p\theta = \cot q\theta$. 3
- (c) Given that A , B and C are the angles of a triangle,
- (a) show that $\tan C = \frac{\tan A + \tan B}{\tan A \tan B - 1}$. 1
- Given further that $\tan B = 3$ and that $\tan C = 2 \tan A$,
- (b) find the angle A without the use of a calculator. 3
- (c) Find all integral solutions of the following simultaneous equations: 4

$$\begin{aligned} x^2 + y &= 3 \dots \boxed{1} \\ y^2 + x &= 5 \dots \boxed{2} \end{aligned}$$

End of Paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x$, $x > 0$

Question 1 (20 marks)

- (a) Express $\frac{2}{3x+6} + \frac{3x+2}{2x^2+x-6}$ as a simplified fraction.

2

Solution:
$$\frac{2}{3(x+2)} + \frac{3x+2}{(x+2)(2x-3)} = \frac{4x-6+9x+6}{3(x+2)(2x-3)},$$

$$= \frac{13x}{3(x+2)(2x-3)}.$$

- (b) Solve the following for x :

(i) $7 - 2x > 12$,

1

Solution:
$$\begin{aligned} -2x &> 5, \\ x &< -5/2. \end{aligned}$$

(ii) $|3x - 2| = 4$,

2

Solution:
$$\begin{aligned} 3x - 2 &= 4, & \text{or} & & -3x + 2 &= 4, \\ 3x &= 6, & & & -3x &= 2, \\ \therefore x &= 2, & & & -2/3. \end{aligned}$$

(iii) $x^{5/3} = 32$,

1

Solution:
$$\begin{aligned} x^{5/3} &= 2^5, \\ x^{1/3} &= 2, \\ \therefore x &= 8. \end{aligned}$$

(iv) $\frac{1}{x} \geq \frac{1}{6}$.

2

Solution: Clearly $x > 0$,
then multiplying both sides by $6x$ gives $6 \geq x$,
so $0 < x \leq 6$.

- (c) If x degrees is equal to X radians, find the value of $\frac{x}{X}$ to two decimal places.

1

Solution:
$$\begin{aligned} x \text{ degrees} &= x \times \frac{\pi}{180} \text{ radians}, \\ x \times \frac{\pi}{180} &= X, \\ \therefore \frac{x}{X} &= \frac{180}{\pi}, \\ &= 57.30. \end{aligned}$$

- (d) If $\tan \frac{\theta}{2} = t$, express $\cot \theta$ in terms of t .

1

Solution:
$$\begin{aligned} \cot \theta &= \frac{1}{\tan \theta}, \\ &= \frac{1}{\frac{2t}{1-t^2}}, \\ &= \frac{1-t^2}{2t}. \end{aligned}$$

(e) The normals to the parabola $x^2 = 4ay$ at the points $P(2at, at^2)$ and $Q(2as, as^2)$ meet at R .

(i) Find the coördinates of R in terms of s and t .

4

Solution: $\frac{dy}{dx} = \frac{2x}{4a},$
 $= \frac{x}{2a},$
 $= t \text{ at } P \text{ and } s \text{ at } Q.$
 Normal at $P: y - at^2 = -\frac{1}{t}(x - 2at),$
 $yt - at^3 = 2at - x,$
 $x + ty = at^3 + 2at \dots \boxed{1}$
 Similarly, at $Q, x + sy = as^3 + 2as \dots \boxed{2}$
 Solving simultaneously: $\boxed{1} - \boxed{2}: y(t - s) = a((t^3 - s^3) + 2(t - s)),$
 $y = a(t^2 + st + s^2 + 2),$
 sub. in $\boxed{1}: x = at^3 + 2at - at^3 - ast^2 - as^2t - 2at,$
 $= -ast(t + s).$
 $\therefore R \text{ is } (-ast(t + s), a(t^2 + st + s^2 + 2)).$

(ii) If $st = -2$, find the cartesian equation of the locus of R .

3

Solution: $x = 2a(t + s),$ $y = a(t^2 + s^2),$
 $t + s = \frac{x}{2a},$ $t^2 + s^2 = \frac{y}{a},$
 $(t + s)^2 - 2ts = \frac{y}{a},$
 $\left(\frac{x}{2a}\right)^2 + 4 = \frac{y}{a},$
 $4ay = x^2 + 16a^2.$

(f) A curve is given parametrically by the equations $x = t^2, y = t^3$, where t is a parameter. Show that the tangent at the point with parameter t has the equation $2y - 3tx + t^3 = 0$.

3

Solution: $\frac{dy}{dx} = \frac{dy}{dt} \div \frac{dx}{dt},$
 $= \frac{3t^2}{2t},$
 $= \frac{3}{2}t.$

At the point with parameter t , the equation of the tangent is

$$y - t^3 = \frac{3}{2}t(x - t^2),$$

$$2y - 2t^3 = 3tx - 3t^3,$$

$$\text{i.e. } 2y - 3tx + t^3 = 0.$$

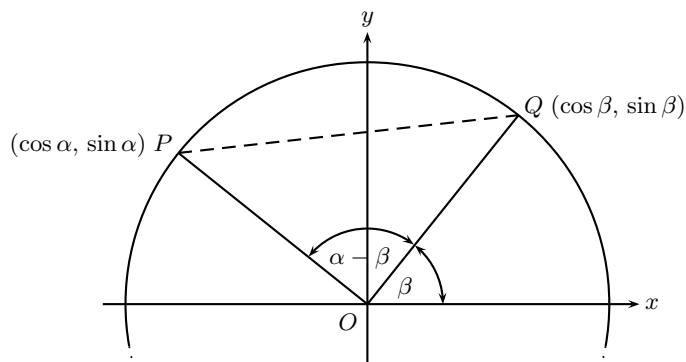
Question 2 (20 marks)

(Use a separate writing booklet.)

- (a) P is the point $(\cos \alpha, \sin \alpha)$ and Q is the point $(\cos \beta, \sin \beta)$ on the circumference of the circle, centre the origin, radius 1. By writing down two different expressions for PQ^2 , deduce the result that

3

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta.$$

Solution:

Using the distance formula:

$$\begin{aligned} PQ^2 &= (\cos \alpha - \cos \beta)^2 + (\sin \alpha - \sin \beta)^2, \\ &= \cos^2 \alpha - 2 \cos \alpha \cos \beta + \cos^2 \beta + \sin^2 \alpha - 2 \sin \alpha \sin \beta + \sin^2 \beta, \\ &= 2 - 2(\cos \alpha \cos \beta + \sin \alpha \sin \beta). \end{aligned}$$

Using the cosine formula:

$$\begin{aligned} PQ^2 &= OP^2 + OQ^2 - 2OP \cdot OQ \cdot \cos(\alpha - \beta), \\ &= 2 - 2 \cos(\alpha - \beta). \end{aligned}$$

Equating these expressions:

$$\begin{aligned} 2 - 2 \cos(\alpha - \beta) &= 2 - 2(\cos \alpha \cos \beta + \sin \alpha \sin \beta), \\ \cos(\alpha - \beta) &= \cos \alpha \cos \beta + \sin \alpha \sin \beta. \end{aligned}$$

- (b) Factorise the expression $x^2 - 6x + 9$, and hence find any real solutions for $x^4 - x^2 + 6x - 9 = 0$.

3

Solution:

$$\begin{aligned} x^2 - 6x + 9 &= (x - 3)^2, \\ \therefore x^4 - x^2 + 6x - 9 &= (x^2)^2 - (x - 3)^2, \\ &= (x^2 - x + 3)(x^2 + x - 3). \end{aligned}$$

Now $x^2 - x + 3 = 0$ has no real solutions as the discriminant, $\Delta = 1 - 4 \times 3, < 0$.

However $x^2 + x - 3 = 0$,

$$\begin{aligned} x^2 + x + 1/4 &= 3 + 1/4, \\ (x + 1/2)^2 &= \frac{13}{4}, \\ x + 1/2 &= \pm \frac{\sqrt{13}}{2}, \\ x &= \frac{-1 \pm \sqrt{13}}{2}. \end{aligned}$$

- (c) The equation $\frac{5}{y^2} - \frac{1}{2y} = 1$ is true when $y = 2$.

2

Use this fact to find a solution of $\frac{5}{(2y+5)^2} - \frac{1}{2(2y+5)} = 1$.

Solution: $2y + 5 = 2,$
 $2y = -3,$
 $y = -3/2.$

- (d) Find the acute angle between the curves $y = x^2$ and $y = 3 - 2x$ at the point where they meet in the first quadrant.
 (Give your answer to the nearest degree.)

3

Solution: Solving simultaneously,

$$\begin{aligned} x^2 &= 3 - 2x, \\ x^2 + 2x - 3 &= 0, \\ (x+3)(x-1) &= 0, \\ x &= -3, 1. \end{aligned}$$

So the intersection in the first quadrant is (1, 1).

For $y = x^2$, $\frac{dy}{dx} = 2x = 2$ at the intersection.

The slope of $y = 3 - 2x$ is -2.

$$\begin{aligned} \tan(\text{angle}) &= \left| \frac{2 - (-2)}{1 + 2 \times -2} \right|, \\ &= 4/3. \\ \therefore \text{angle} &= 53.13010235 \dots \\ &\approx 53^\circ. \end{aligned}$$

- (e) If α , β and γ are the roots of the equation $x^3 + 2x^2 + 3x + 4 = 0$, find the value of
 (i) $(\alpha - 1)(\beta - 1)(\gamma - 1)$,

3

Solution: We note first that $\alpha + \beta + \gamma = -2$,
 $\alpha\beta + \alpha\gamma + \beta\gamma = 3$,
 and $\alpha\beta\gamma = -4$.

$$\begin{aligned} \text{Then } (\alpha - 1)(\beta - 1)(\gamma - 1) &= \alpha\beta\gamma - (\alpha\beta + \alpha\gamma + \beta\gamma) + (\alpha + \beta + \gamma) - 1, \\ &= -4 - 3 - 2 - 1, \\ &= -10. \end{aligned}$$

- (ii) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$,

1

Solution: $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\alpha\beta + \alpha\gamma + \beta\gamma}{\alpha\beta\gamma},$
 $= -\frac{3}{4}.$

- (iii) $\alpha^2 + \beta^2 + \gamma^2$.

1

Solution: $(\alpha + \beta + \gamma)^2 = \alpha^2 + \beta^2 + \gamma^2 + 2(\alpha\beta + \alpha\gamma + \beta\gamma),$
 So $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma),$
 $= (-2)^2 - 2 \times 3,$
 $= -2.$

- (f) (i) Find the value of n for which the division of $3x^n - 7x^2 + 15$ by $x - 2$ leaves a remainder of 11.

2

Solution: Put $P(x) = 3x^n - 7x^2 + 15$,
then $P(2) = 3 \times 2^n - 7 \times 4 + 15 = 11$,
 $3 \times 2^n = 24$,
 $2^n = 8$,
 $\therefore n = 3$.

- (ii) Given that $16x^4 - 4x^3 - 4b^2x^2 + 7bx + 18$ is divisible by $2x + b$, show that $b^3 - 7b^2 + 36 = 0$.

2

Solution: Put $P(x) = 16x^4 - 4x^3 - 4b^2x^2 + 7bx + 18$,
then $P\left(-\frac{b}{2}\right) = 0 = \frac{16b^4}{16} + \frac{4b^3}{8} - 4b^2 \times \frac{b^2}{4} - \frac{7b^2}{2} + 18$,
 $= b^4 + \frac{b^3}{2} - b^4 - \frac{7b^2}{2} + 18$,
 $= \frac{b^3}{2} - \frac{7b^2}{2} + 18$,
i.e. $b^3 - 7b^2 + 36 = 0$.

Question 3 (20 marks)

(Use a separate writing booklet.)

- (a) The points $P(-5, 1)$, $A(a, -3)$ and $B(-2, b)$ are such that P divides the interval AB externally in the ratio $4 : 1$. Find the values of a and b .

3

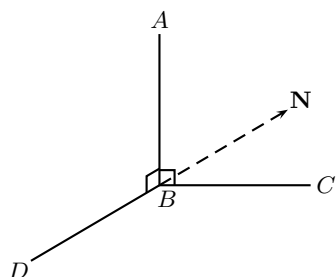
Solution: $(-5, 1) = \left(\frac{-1 \times a + 4 \times -2}{-1 + 4}, \frac{-1 \times -3 + 4 \times b}{-1 + 4} \right),$
i.e. $-a - 8 = -15,$
 $a = 7,$
 and $3 + 4b = 3,$
 $b = 0.$

- (b) Solve $(x + \frac{1}{x})^2 + 6 = 5(x + \frac{1}{x})$ for x .

3

Solution: Put $y = (x + \frac{1}{x}).$
 $y^2 - 5y + 6 = 0,$
 $(y - 3)(y - 2) = 0,$
 $y = 2, 3.$
 $\therefore x + \frac{1}{x} = 2, \text{ or } x + \frac{1}{x} = 3,$
 $x^2 - 2x + 1 = 0, \quad x^2 - 3x + 1 = 0,$
 $(x - 1)^2 = 0, \quad x = \frac{3 \pm \sqrt{9 - 4}}{2},$
i.e. $x = 1 \text{ or } x = \frac{3 \pm \sqrt{5}}{2}.$

- (c)



B, C and D are three points on a horizontal plane.

C is due east of B and D is due south of B .

C and D are 400 metres apart.

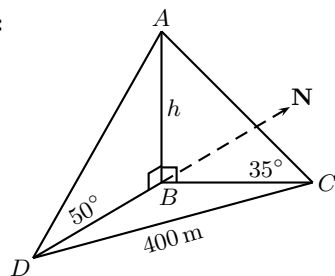
A is the top of a vertical tower AB .

From C the angle of elevation of A is 35° and from D the angle of elevation of A is 50° .

The height of the tower is h metres.

- (i) Copy the diagram and mark on it the given distances and angles.

1

Solution:

- (ii) Show that $\frac{h^2}{\tan^2 50^\circ} + \frac{h^2}{\tan^2 35^\circ} = 400^2$.

2

Solution: $\tan 50^\circ = \frac{h}{DB}, \quad \tan 35^\circ = \frac{h}{BC},$
 $DB = \frac{h}{\tan 50^\circ}, \quad BC = \frac{h}{\tan 35^\circ}.$
 Now $DB^2 + BC^2 = 400^2,$
 $\therefore \frac{h^2}{\tan^2 50^\circ} + \frac{h^2}{\tan^2 35^\circ} = 400^2.$

- (iii) Find h , correct to one decimal place.

1

Solution: $h^2 \left(\frac{1}{\tan^2 50^\circ} + \frac{1}{\tan^2 35^\circ} \right) = 400^2,$
 $h^2 = 400^2 \times \left(\frac{\tan^2 50^\circ \tan^2 35^\circ}{\tan^2 50^\circ + \tan^2 35^\circ} \right),$
 $h \approx 400 \times \sqrt{0.36447201},$
 $h \approx 241.5 \text{ m}.$

- (iv) Find the bearing of D from C to the nearest degree.

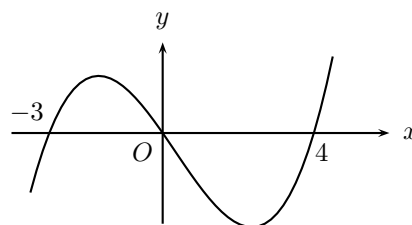
2

Solution: $\tan \widehat{BCD} = \frac{BD}{BC},$
 $= \frac{\frac{h}{\tan 50^\circ}}{\frac{h}{\tan 35^\circ}} \div \frac{h}{\tan 35^\circ},$
 $\approx 0.58754389.$
 $\therefore \widehat{BCD} \approx 30.4361^\circ.$
 So the bearing of D from C is 240° .

- (d) (i) Find the set of values of x for which $\frac{x^2 - 12}{x} > 1$.

3

Solution: $\frac{x^2 - 12}{x} - 1 > 0,$
 $\frac{x^2 - 12 - x}{x} > 0,$
 $\frac{(x - 4)(x + 3)}{x} > 0,$
 $x(x - 4)(x + 3) > 0.$



$\therefore -3 < x < 0, x > 4.$

- (ii) Deduce the set of values of x for which $\frac{x^2 - 12}{|x|} > 1$.

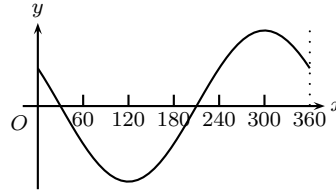
1

Solution: Clearly, when $-3 < x < 0, x^2 - 12 < 0.$
 $\therefore |x| > 4.$

- (e) (i) For what values of x between 0 and 360 is $\cos x^\circ - \sqrt{3} \sin x^\circ$ positive?

2

Solution: $r = \sqrt{1+3} = 2,$
 $\alpha = \tan^{-1} \sqrt{3} = 60^\circ,$
 $2 \cos(x + 60)^\circ > 0,$
 $\therefore 0 \leq x < 30, 210 < x \leq 360.$



- (ii) Find all values of ψ for which $\tan^2 \psi = \tan \psi$.

2

Solution: $\tan \psi (\tan \psi - 1) = 0,$
 $\tan \psi = 0$ or $1,$
 $\psi = n\pi$ or $n\pi + \frac{\pi}{4}.$

Question 4 (20 marks)

(Use a separate writing booklet.)

- (a) Consider the curve expressed parametrically as
- $x = t + 1$
- ,
- $y = 2t^2 - 2$
- .

- (i) Re-express this relationship in Cartesian form,
- i.e.*
- , as
- $y = f(x)$
- .
- 1

Solution:

$$\begin{aligned} t &= x - 1, \\ y &= 2(x - 1)^2 - 2, \\ y &= 2x^2 - 4x + 2 - 2, \\ y &= 2x^2 - 4x. \end{aligned}$$

- (ii) Find its focus and directrix.
- 2

Solution: $y + 2 = 4 \times \frac{1}{2}(x - 1)^2$.
 $\therefore$ Vertex is at $(1, -2)$ and focal length is $\frac{1}{2}$.
 Thus the focus is $(1, -1\frac{1}{2})$ and the directrix is the line $y = -2\frac{1}{2}$.

- (iii) Derive the equation of the tangent at the point
- $A(x_1, y_1)$
- on the curve.
- 2

Solution: $\frac{dy}{dx} = 4x - 4$,
 $= 4(x_1 - 1)$ at (x_1, y_1) .
 So the equation is $y - y_1 = 4(x_1 - 1)(x - x_1)$.

- (iv) Hence or otherwise, show that the chord of contact of the tangents from an external point
- $T(x_0, y_0)$
- is
- $y + y_0 = 4(xx_0 - x - x_0)$
- .
- 4

Solution: Similarly, the tangent at $B(x_2, y_2)$ is $y - y_2 = 4(x_2 - 1)(x - x_2)$,
 $y - y_2 = 4(xx_2 - x_2^2 - x + x_2)$,
 but (x_2, y_2) lies on the curve so $2x_2^2 = 2(2xx_2 - y_2 - 4x_2 - 2x + 2x_2)$ and
 $y - y_2 = 2(2xx_2 - y_2 - 4x_2 - 2x + 2x_2)$,
 $y = 4xx_2 - y_2 - 4x_2 - 4x$,
 $y + y_2 = 4(xx_2 - x_2 - x)$.
 Now as the point $T(x_0, y_0)$ lies at the intersection of both tangents,
 $y_0 + y_1 = 4(x_0x_1 - x_1 - x_0)$,
 $y_0 + y_2 = 4(x_0x_2 - x_2 - x_0)$.
 and as $A(x_1, y_1)$ and $B(x_2, y_2)$ both lie on the chord of contact, we deduce that
 $y_0 + y = 4(xx_0 - x - x_0)$,
i.e. $y + y_0 = 4(xx_0 - x - x_0)$ is the desired equation.

- (b) Find the general solution to
- $\tan p\theta = \cot q\theta$
- .
- 3

Solution: $\tan p\theta \times \tan q\theta = 1$,
 $\frac{\sin p\theta}{\cos p\theta} \times \frac{\sin q\theta}{\cos q\theta} = 1$,
 $\sin p\theta \sin q\theta = \cos p\theta \cos q\theta$,
 $\cos p\theta \cos q\theta - \sin p\theta \sin q\theta = 0$,
 $\cos(p + q)\theta = 0$,
 $\cos(p + q)\theta = \cos \frac{(2n + 1)\pi}{2}$, $n \in \mathbb{J}$,
 $(p + q)\theta = \frac{(2n + 1)\pi}{2}$,
 $\therefore \theta = \frac{(2n + 1)\pi}{2(p + q)}$.

(c) Given that A , B and C are the angles of a triangle,

(a) show that $\tan C = \frac{\tan A + \tan B}{\tan A \tan B - 1}$.

1

Solution: $\tan C = \tan(\pi - \overline{A + B}),$
 $= -\tan(A + B),$
 $= -\frac{\tan A + \tan B}{1 - \tan A \tan B},$
 $= \frac{\tan A + \tan B}{\tan A \tan B - 1}.$

Given further that $\tan B = 3$ and that $\tan C = 2 \tan A$,

(b) find the angle A without the use of a calculator.

3

Solution: $2 \tan A = \frac{\tan A + 3}{3 \tan A - 1}.$
Put $\tan A = x$, then $2x = \frac{x + 3}{3x - 1},$
 $6x^2 - 2x = x + 3,$
 $6x^2 - 3x - 3 = 0,$
 $2x^2 - x - 1 = 0,$
 $(2x + 1)(x - 1) = 0,$
 $\therefore \tan A = 1 \text{ or } -1/2.$
 $\implies A = \frac{\pi}{4} \text{ or } A > \frac{\pi}{2}, \text{ and } \tan C = -1 \text{ so } C = \frac{3\pi}{4}.$
But we can't have both A and C greater than $\frac{\pi}{2}$, so the only solution is $A = \frac{\pi}{4}.$

(c) Find all integral solutions of the following simultaneous equations:

4

$$\begin{aligned}x^2 + y &= 3 \dots \boxed{1} \\ y^2 + x &= 5 \dots \boxed{2}\end{aligned}$$

Solution: From $\boxed{1}$: $y = 3 - x^2$.

Sub. in $\boxed{2}$: $(3 - x^2)^2 + x = 5$,

$$x^4 - 6x^2 + x + 4 = 0.$$

Now, putting $P(x) = x^4 - 6x^2 + x + 4$,

$$P(1) = 1 - 6 + 1 + 4 = 0.$$

So $(x - 1)$ is a factor of $P(x)$ and, using long division,

$$\begin{array}{r}x^3 + x^2 - 5x - 4 \\x - 1 \overline{) \begin{array}{r} x^4 - 6x^2 + x + 4 \\ - x^4 + x^3 \\ \hline x^3 - 6x^2 + x + 4 \\ - x^3 + x^2 \\ \hline - 5x^2 + x + 4 \\ 5x^2 - 5x \\ \hline - 4x + 4 \\ 4x - 4 \\ \hline 0 \end{array}}\end{array}$$

or, for those who prefer Horner's scheme,

$$\begin{array}{r|rrrrr}1 & 1 & 0 & -6 & 1 & 4 \\ & & 1 & 1 & -5 & -4 \\ \hline & 1 & 1 & -5 & -4 & 0\end{array}$$

Now, putting $Q(x) = x^3 + x^2 - 5x - 4$,

$$Q(1) = 1 + 1 - 5 - 4 \neq 0,$$

$$Q(-1) = -1 + 1 + 5 - 4 \neq 0,$$

$$Q(2) = 8 + 4 - 10 - 4 \neq 0,$$

$$Q(-2) = -8 + 4 + 10 - 4 \neq 0.$$

So there are no more integral solutions.

When $x = 1$, $y = 2$; so the only integral solution is $(1, 2)$.

End of Paper